

Multigrid Finite State Projection for the Reaction-Diffusion Master Equation

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The Chemical Master Equation: exact stochastic distributions

A well-mixed system: integer counts $\mathbf{n} \in \mathbb{Z}_{\geq 0}^S$ evolving by stochastic reactions.

The probability $p(\mathbf{n}, t)$ satisfies the **CME**:

$$\dot{p} = A p,$$

where A encodes every reaction event as a transition rate.

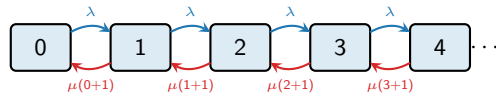
SSA vs. CME

SSA samples individual trajectories.

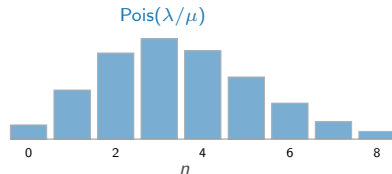
CME gives the *exact* joint distribution $p(\mathbf{n}, t)$ directly — rare events, tails, and multimodal structure without ensemble noise.

FSP: truncate A to a finite support and solve $\dot{p} = A p$

Birth-death: $\emptyset \xrightarrow{\lambda} X \xrightarrow{\mu n} \emptyset$



Stationary distribution: $\text{Poisson}(\lambda/\mu)$.



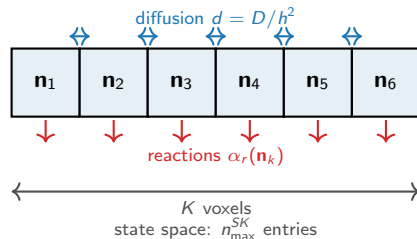
The CME is exact. SSA samples this distribution; FSP computes it directly.

The RDME: reactions and diffusion across a spatial grid

Spatial extension of the CME. Divide a 1-D domain into K voxels. Each voxel k holds molecule counts $\mathbf{n}_k \in \mathbb{Z}_{\geq 0}^S$.

Two types of events:

- **Reactions** inside voxel k : rate $\alpha_r(\mathbf{n}_k)$, change $\mathbf{n}_k$.
- **Diffusion** between neighbors: rate $d = D/h^2$, moves one molecule $k \leftrightarrow k+1$.

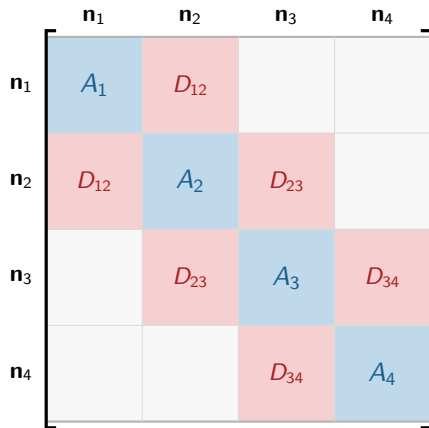


The joint probability $p(\mathbf{n}_1, \dots, \mathbf{n}_K, t)$ satisfies $\dot{p} = Ap$ (**RDME**).

The computational bottleneck

The joint state space has $\sim n_{\max}^{SK}$ entries — **exponential in K** . Even $K=8$, $S=1$ is intractable with a naive FSP.

$\dot{p} = Ap$: **block-tridiagonal structure** ($K=4$, $S=1$)



A_k : reactions inside voxel k (dense block)

$D_{k,k+1}$: diffusion $k \leftrightarrow k+1$ (sparse: shifts one count)

Key locality

Diffusion only couples *adjacent* blocks.

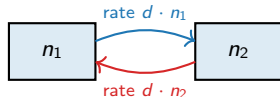
Merging columns $(k, k+1)$ into a single coarse voxel $\bar{k}$ is **exact**: $D_{k,k+1}$ becomes internal to $\bar{k}$ and drops out of the inter-block coupling.

This is the foundation of spatial coarsening:
restriction $\mathcal{R}$ sums each adjacent pair exactly;
the coarse generator is exact up to the within-pair distribution π .

The Binomial is the exact diffusion equilibrium

Fix total molecules $\bar{n} = n_1 + n_2$ across two adjacent voxels.

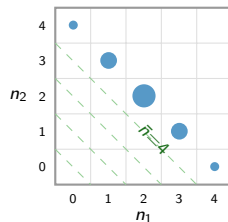
Diffusion moves one molecule at a time, at rate d per molecule:



Each molecule hops independently $\Rightarrow$ at equilibrium, equally likely to be in either voxel. For $\bar{n}$ total:

$$P(n_1 \mid \bar{n}) = \text{Bin}(\bar{n}, \frac{1}{2}) \quad (\text{exact, any } d > 0)$$

When the distribution is near this equilibrium, $\bar{n}$ **alone describes the two-voxel state.**



Probability spreads along constant-total diagonals as $\text{Bin}(\bar{n}, \frac{1}{2})$ (dot size $\propto$ mass).

The two-voxel state is captured by $\bar{n}$ alone.

Merging two adjacent voxels: restriction and prolongation

Compress ($\mathcal{R}$): exact:

$$p^c(\bar{n}) = \sum_{n_1+n_2=\bar{n}} p(n_1, n_2).$$

Reconstruct ($\mathcal{P}$): approximate:

$$p(n_1, n_2) \approx p^c(\bar{n}) \cdot \pi(\bar{n}),$$

where π is the **within-pair distribution**.

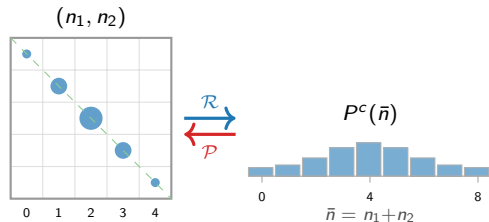
State-space reduction

Per pair: $n_{\max}^{2S} \rightarrow n_{\max}^S$.

All K voxels: $n_{\max}^{SK} \rightarrow n_{\max}^{SK/2}$.

$\pi = \text{Bin}(\bar{n}, \frac{1}{2})$: exact when diffusion $\gg$ reactions
(within-pair distribution at equilibrium).

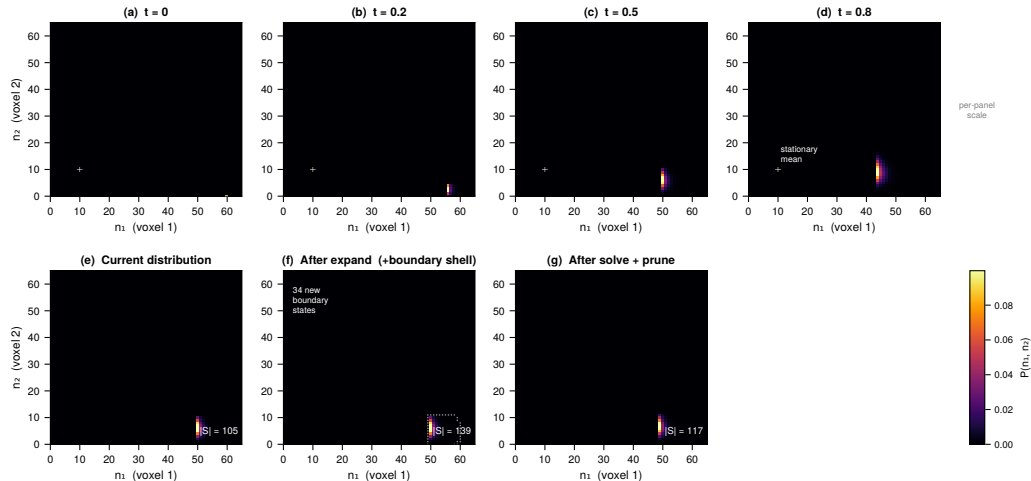
Birth-death: this is essentially exact.



$\mathcal{R}$: sum along each diagonal $\rightarrow$ scalar $\bar{n} = n_1 + n_2$.

$\mathcal{P}$: spread back via within-pair distribution π .

The adaptive FSP algorithm: expand, solve, prune



Top row: joint distribution $P(n_1, n_2)$ of a 2-voxel birth-death system: point source at $(n_{src}, 0)$ traveling to the stationary point (white cross).

The V-cycle: multigrid FSP

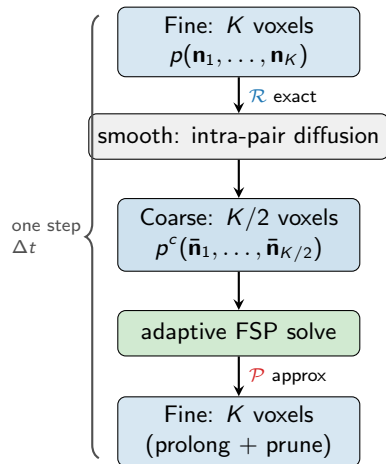
One V-cycle ($K \rightarrow K/2 \rightarrow K$):

- 1 **Pre-smooth**: intra-pair diffusion for τ_{pre}
- 2 **Restrict**: $\bar{n}_j = n_{2j-1} + n_{2j}$ (exact)
- 3 **Coarse FSP**: adaptive solve on $K/2$ voxels
- 4 **Prolong**: reconstruct fine distribution
- 5 **Prune**: drop states with $p(x) < \tau$

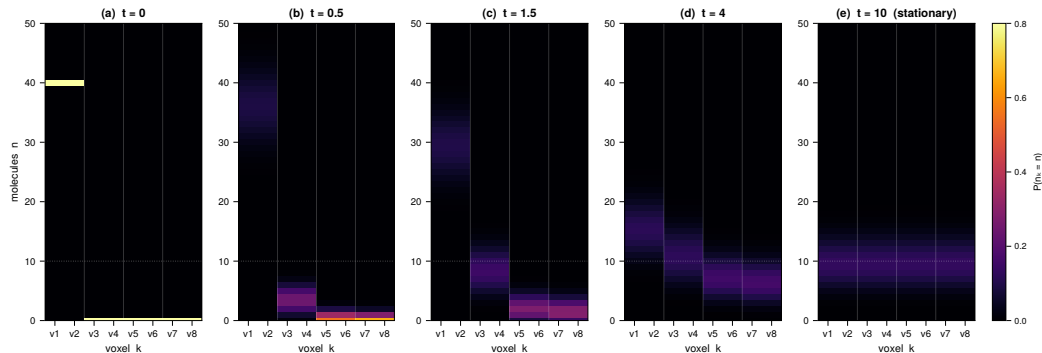
Prolongation for covered states:

$$p^{\text{new}}(x) = \tilde{p}(x) \cdot \frac{\bar{p}^{\text{post}}(\bar{x})}{\bar{p}^{\text{pre}}(\bar{x})}$$

Multiplicative update — exact for birth-death;
preserves bistable orientation for Schlögl.



Result: K=8 RDME wavefront via coarsened simulation



K=8 wavefront solved via K=4 coarse FSP. Max coarse state space: 106585 states. K=8 direct FSP would be intractable (grows to 90k+ states by $t=0.8$). Fine marginals recovered analytically via $\text{Binom}(n_k, \frac{1}{2})$ prolongation — exact for balanced pairs.

Birth-death RDME, $K=8$, $D=0.1$. IC: $v_1=v_2=40$ (delta), $v_3 \dots v_8=0$. Each panel shows marginals $P(n_k=n)$ for all 8 voxels.

(a)–(d): Wavefront sweeps left→right, distributions broaden. **(e):** Stationary: all voxels reach $\text{Poisson}(10)$ (dashed line).

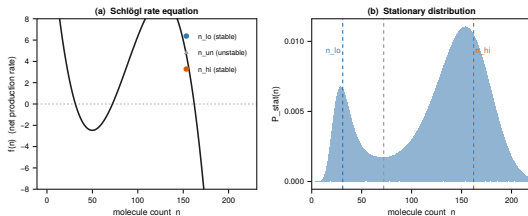
Solved via **K=4 coarse FSP** (30k states). Direct K=8 FSP: intractable. Fine marginals recovered via $\text{Bin}(\tilde{n}, \frac{1}{2})$:

Bistable reactions

For birth-death, each voxel has one stable state.
Diffusion wins: the pair is always well-mixed.
Binomial coarsening is *essentially exact*.

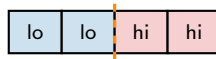
But what if reactions are bistable?

The **Schlögl model** has two stable states per voxel:
 $n_{lo} \approx 31$ and $n_{hi} \approx 162$.



Spatial metastability

Neighbors can occupy *different* states. A **spatial front** forms.



Reactions “pull” voxels apart; diffusion “pulls” them together. When reactions are fast, the front **persists for very long times**.

Why Binomial prolongation fails for bistable fronts

At the spatial front, neighboring voxels occupy *different* stable states. The joint distribution $P(n_1, n_2)$ is **bimodal**: mass at (n_{lo}, n_{hi}) and (n_{hi}, n_{lo}) .

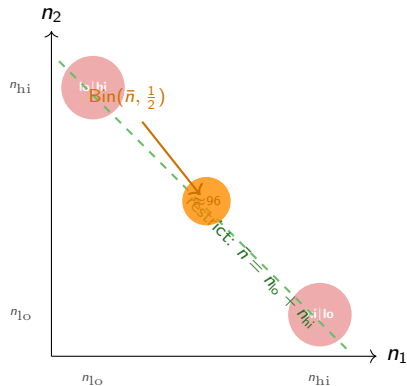
The symmetry trap

Restriction maps both front orientations to the same scalar:

$$\mathcal{R}(n_{lo}, n_{hi}) = \bar{n} = \mathcal{R}(n_{hi}, n_{lo}).$$

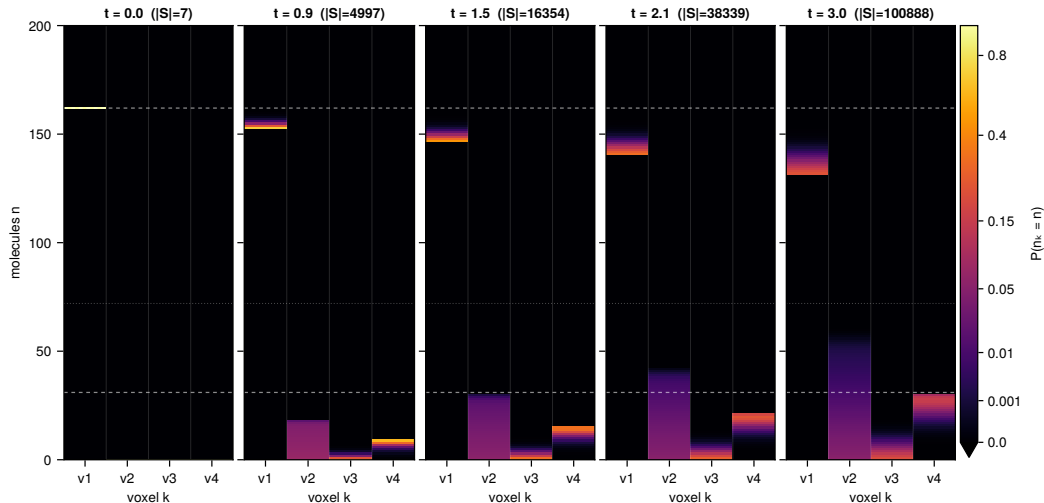
Front direction is erased.

Binomial prolongation places the reconstructed mass at the **unstable midpoint** ≈ 96 — the system immediately leaves this state.



Result: Schlögl wavefront spreading via the V-cycle

Schlögl RDME: $K=4$, $D=1.0$ — IC: all 162 molecules in v_1 , spreading via diffusion [color: $P^{(1/3)}$]



Summary

1. Restriction + prolongation compresses the RDME state space.

Marginalizing adjacent-voxel pairs reduces the state space from $n_{\max}^{SK}$ to $n_{\max}^{SK/2}$ per V-cycle level. Binomial prolongation is exact in the fast-diffusion limit.

K=8 birth-death wavefront solved via K=4 coarse FSP (30k states; direct: intractable).

2. Conditional prolongation handles bistable fronts.

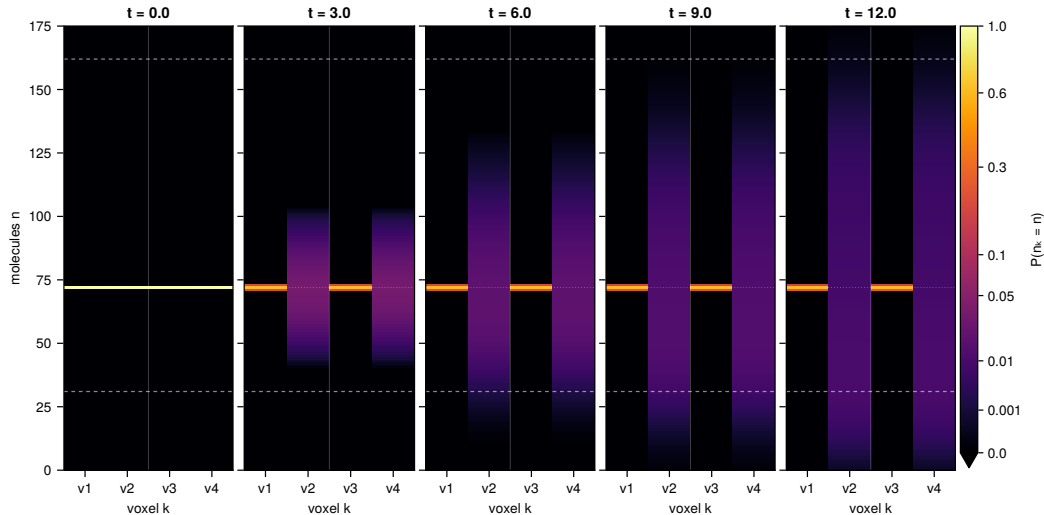
Multiplicative update for covered states preserves within-pair asymmetry — mass at $(n_{\text{lo}}, n_{\text{hi}})$ is never symmetrised to the unstable midpoint.

$L_1 \leq 2$ neighbor search correctly seeds new states at both reaction and diffusion boundaries, propagating the front across pairs.

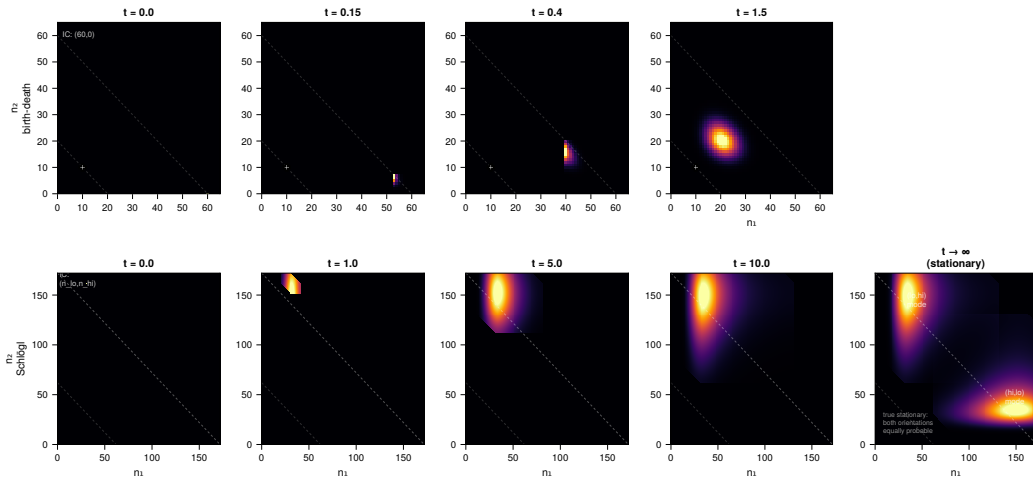
Schlögl wavefront (K=4, controlled state-space growth) correctly captured.

Appendix: Schlögl splitting from the unstable fixed point

Schlögl RDME: $K=4$ voxels, $D=5.0$ — unstable IC ($n=72$) splitting toward low ($n=31$) and high ($n=162$) [color: $P(n_k=n)^{1/3}$]



Appendix: probability dynamics (birth-death vs. Schlögl)



Top: birth-death point source traveling to Poisson(10) stationary. Binomial coarsening exact.

Bottom: Schlögl front IC persisting metastably; true stationary (rightmost) shows both orientations equally.

Appendix: K=8 persistent front

